

HAT-P-12b

R-0863

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_230617 · created 2026-07-28T22:24:17+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460112.7273 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.104 +/- 0.028
Transit depth (Rp/Rs)^2	1.09 +/- 0.58 [%]
Orbital Inclination (inc)	89.96 +/- 1.24
Airmass coefficient 1 (a1)	0.986 +/- 0.015
Airmass coefficient 2 (a2)	0.014 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.53 %
Best Comparison Star	#1 - [121, 221]
Optimal Method	PSF photometry
Transit Duration (day)	0.0936 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.104 ± 0.028 vs 0.1406 (deviation 1.3σ)
Duration fit vs published	0.0936 d vs 0.0975 d (deviation 0.81σ)
Mid-time O-C	-10.7 min
Depth SNR	3.7
Residual scatter	1.53 %

Light curve

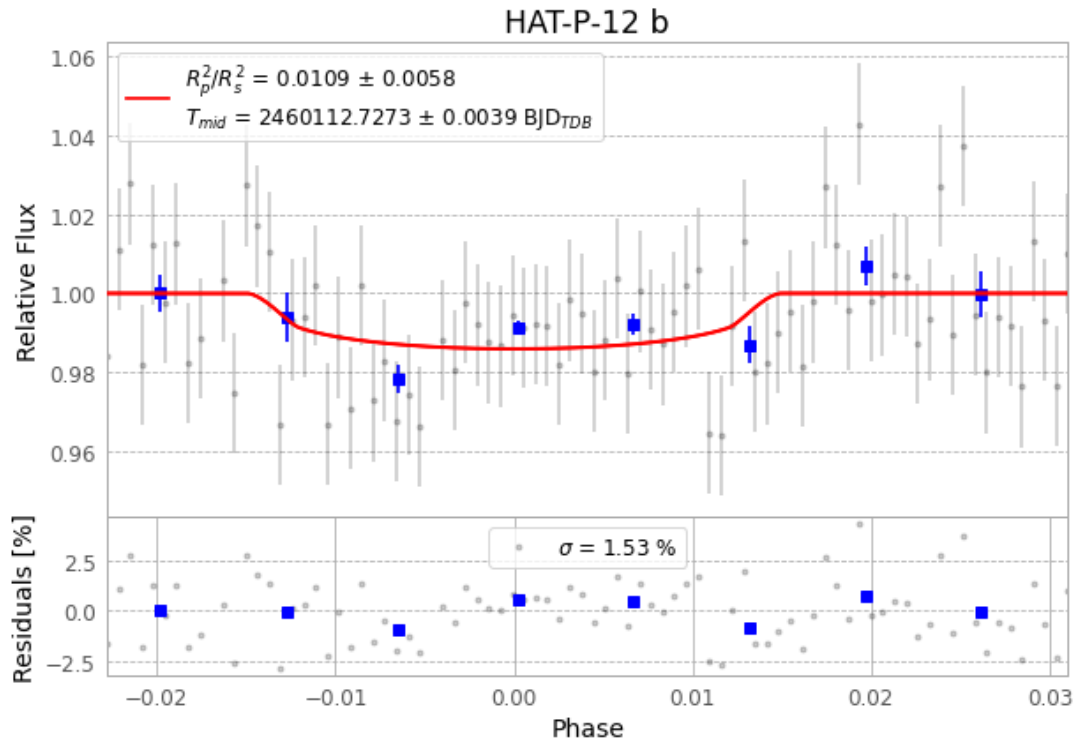


Figure 1 — FinalLightCurve_HAT-P-12 b_2023-06-16.png (SHA-256 cb3d90492c247f46...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [269, 179], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a21ca158fe723721...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

84 FITS frames (55 MB), HATP-12230617033915.FITS → HATP-12230617074815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-230617.log (b4f32b9f732e...), FinalLightCurve_HAT-P-12 b_2023-06-16.png (cb3d90492c24...), FinalLightCurve_HAT-P-12 b_2023-06-16.csv (1b396357c335...)

Compute resources

Wall time 110.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 22:24Z.